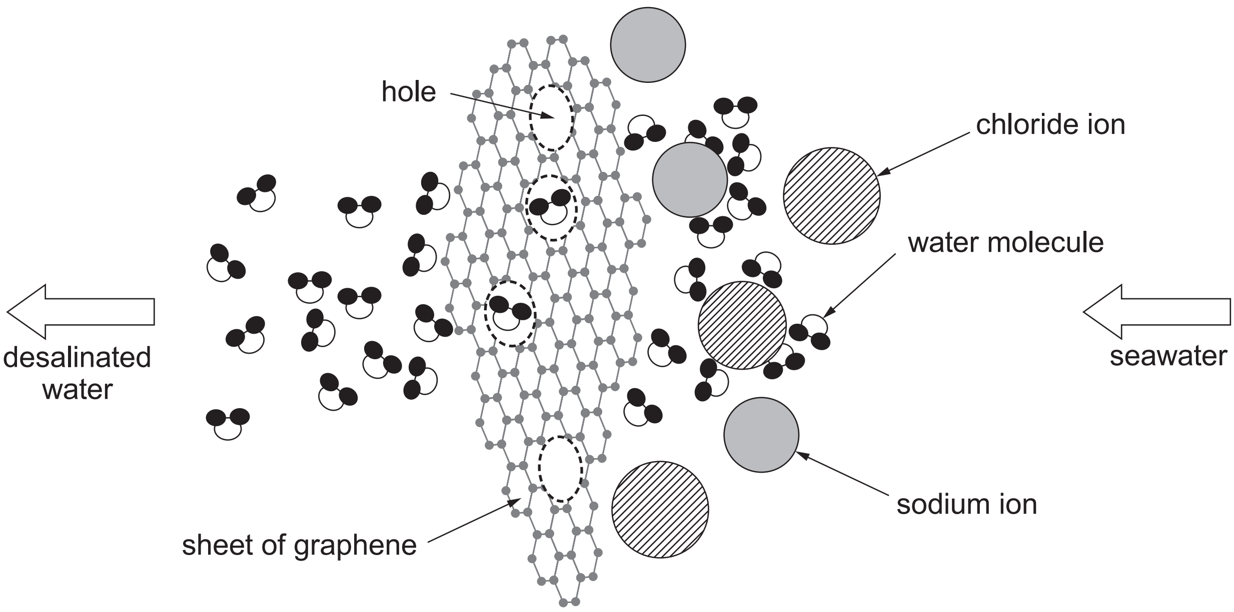
3. *(a)* One out of six people today do not have access to safe fresh water. Estimates suggest fresh water supplies will be a major problem for half the countries of the world by 2025 and by 2050 about 75 % of the world's population will experience a serious scarcity of the resource.

As the world's population increases not only will the demand for drinking water and water for sanitation increase, but the demand for water to produce more food will increase. As climate change makes rainfall less predictable and droughts more common, a growing number of countries are turning to desalination.

One method of desalination, called reverse osmosis, uses a thick membrane to filter salt from seawater. But this system requires an extremely high pressure to force water through the membrane and therefore uses a lot of energy.

Researchers in the USA have come up with a new approach using a different kind of filtration material - sheets of graphene - a one-atom-thick form of the element carbon. This new material can be far more efficient and possibly less expensive than existing desalination methods. Sheets of graphene are perforated with precisely-sized holes which only let the water molecules through.



Examiner only

(i) Tick (✓) the box next to the **main** reason why countries are building desalination plants. [1]

water is becoming scarce ☐

each person is drinking more water ☐

climate change affects the availability of drinking water ☐

(ii) Tick (✓) the statement which **best** explains how graphene sheets desalinate water. [1]

water molecules contain atoms and not ions ☐

sheets contain holes which are bigger than ions ☐

sheets contain holes which are bigger than water molecules but smaller than sodium and chloride ions ☐

sheets contain holes which let molecules through but not ions ☐

molecules are smaller than ions ☐

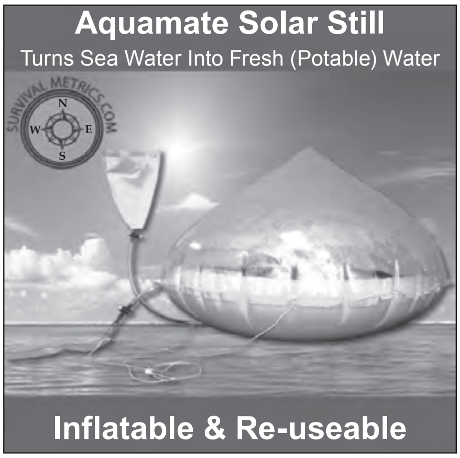
(b) The total volume of water on the Earth is 1.4×10^9 km^3, 97 % of which is seawater. Most of the remaining 3 % is bound up in ice at the poles, leaving 0.3 % available as fresh water.

Calculate the volume of the Earth's available fresh water. Give your answer in standard form. [2]

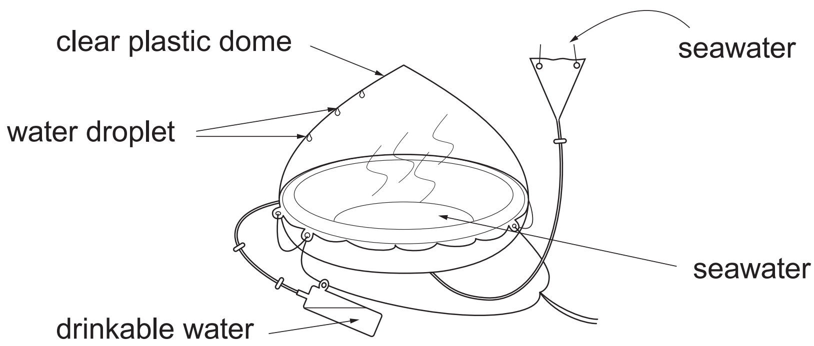
volume of available fresh water = ... km^3

Examiner only

(c) One piece of survival equipment found on life rafts is a 'solar still'. Solar stills are used to remove soluble salts from seawater to form drinkable water.



Source: *www.survivalmetrics.com*



State and explain the physical changes that occur in the formation of drinkable water from seawater. Include the name given to the overall process taking place. [4]

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